

Supplementary Information

Unconcentrated markets, captive buyers: incumbency in Türkiye’s public IT procurement

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S1 Data construction

S1.1 Source and IT scope

Award records were collected from EKAP (ekapv2.kik.gov.tr) by searching domain keywords. Search matches substrings of tender titles, which admits unrelated tenders. The keyword *erp* retrieved 618 contracts, of which 546 are not IT (it matches, among others, the place name Yaverpaşa and *tuz serpici*, a salt spreader). The phrase *teknik destek* retrieved 1,186 contracts, of which 798 are hospital or university staffing and cleaning contracts. An earlier count-based cleaning had removed 394 obvious false positives from 13,418 records, leaving 13,024.

We re-audited this set by value. All 150 contracts with the highest value were classified by hand (108 IT, 1 call-centre service, 19 mixed, 22 not IT). For the remainder, ordered title rules assign each contract to one of four scope classes: IT, IT-enabled call-centre services, ambiguous, and not IT. Works contracts are ambiguous only if they install an IT or operational-technology system, such as city security cameras, SCADA, fibre, or a data centre, and are otherwise not IT. Staffing contracts for data entry are ambiguous; bundled cleaning or catering contracts without an IT system are not IT. Random samples from every rule were inspected, and seven rule outcomes for top contracts were overridden by hand. Table S1 reports the result. The main sample consists of the IT and call-centre classes.

S1.2 Firms, buyers, and markets

Firms. Names were canonicalized by upper-casing with Turkish locale, removing punctuation and whitespace, and stripping legal-form suffixes from the end of the name, subject to a minimum remaining length. Joint ventures were kept separate from their members. The merge groups of

Table S1: Scope classes after the value-based audit (13,024 contracts that passed the earlier count-based cleaning).

Scope class	Contracts	Nominal value (bn TRY)	Share of value (%)
IT	9,981	25.64	56.5
IT-enabled call-centre services	10	7.62	16.8
Ambiguous (excluded from main sample)	1,406	4.12	9.1
Not IT (removed)	1,627	8.03	17.7
Total	13,024	45.42	100.0

the roughly 300 largest firms were reviewed by hand. Seventy-two brand groups were confirmed, including firms that changed legal form (for example from limited to joint-stock company) or name while keeping the same brand in non-overlapping years. Three uncertain brand pairs were left unmerged. The process reduces 4,788 recorded names to 4,389 firms. About 536 winners (12.2% of firms, 6.1% of contracts) are natural persons, identified by a name pattern without legal-form or business words; they are pseudonymized in the released data.

Buyers. Thirty-six generic buyer labels pool units from two or more provinces, among them the provincial health directorates and hospital unions of the Ministry of Health. We split them by the province recorded for each tender (1,001 contracts), which turns 3,453 labels into 3,663 buyer nodes; six national bodies with offices in two provinces were not split. The recorded labels are used as a sensitivity check throughout.

Buyer sectors. Buyers were assigned to 11 sectors by ordered keyword rules on the buyer name. Relative to an earlier coding, 1,423 contracts changed sector, mainly because upper-casing without Turkish locale had prevented mixed-case names from matching, and because several rules matched unintended substrings.

Product markets. Contracts were assigned to 14 product markets by ordered rules on the tender title, most specific first. In the main sample, 97% of contracts receive a specific market.

Dates, procedures, values. Dates are tender dates joined by tender identifier (coverage 100%, October 2010 to March 2026). Procedures are open, restricted, and the negotiated procedures of Article 21(a)–(f); the extract contains no direct procurement under Article 22. Values are deflated to 2025 lira with the annual average of the consumer price index (2003=100; 2026 from the linked monthly series).

S2 Concentration

Table S2 reports supplier concentration by product market. Bootstrap intervals are bias-shifted percentile intervals (1,000 resamples of contracts within market). “No max” removes the market’s largest contract; “winsor.” caps contract values at the 99th percentile of the main sample. Table S3 reports medians of HHIs computed within calendar years for market-years with at least 30 contracts; the bias-corrected count index is the unbiased estimator $10,000 \sum_i n_i(n_i - 1) / [n(n - 1)]$, where n_i is firm i ’s number of contracts and n the market-year total.

Decomposition. The pooled index satisfies $H_{\text{agg}} = \sum_m w_m^2 H_m + C$, where w_m is market m ’s share and $C = \sum_i \sum_{m \neq m'} s_{im} s_{im'} w_m w_{m'}$ collects the terms of firms active in several markets. By count, $H_{\text{agg}} = 38.6$, $\sum_m w_m^2 H_m = 26.7$ and $C = 11.9$; $\sum_m w_m^2 = 0.117$ and the demand-weighted mean of market HHIs is 179. By real value the corresponding figures are 148.6, 127.9 and 20.7, with a weighted mean of 1,278 (534 without the largest contract). A quarter of firms are active in more than one market; they hold 69% of contracts.

Table S2: Supplier concentration by product market (main sample; HHI on 0–10,000).

Product market	Contracts	Firms	HHI count [95% CI]	HHI value [95% CI]	Value, no max	Value, winsor.	CR4 value (%)
Health information systems	2,189	205	431 [405, 458]	831 [692, 985]	868	811	53
ERP / management software	1,515	599	84 [71, 97]	132 [73, 219]	120	96	16
Custom software / web / mobile	1,117	734	25 [22, 29]	264 [41, 744]	176	110	27
Software licences	1,047	375	103 [89, 118]	275 [188, 441]	276	270	25
Network / data-centre infrastructure	748	390	72 [60, 87]	281 [151, 523]	234	226	28
Computers / peripherals	696	467	40 [34, 47]	170 [69, 363]	146	155	18
Maintenance / support services	626	297	82 [69, 98]	301 [196, 481]	303	296	25
GIS / city information	553	203	353 [273, 444]	408 [228, 842]	385	347	33
Cybersecurity	390	194	193 [143, 263]	476 [193, 1,002]	442	340	37
Physical security / surveillance	385	222	84 [69, 104]	212 [123, 373]	208	210	20
Other IT	277	215	79 [58, 111]	767 [0, 2,275]	603	296	46
Education technology	205	128	225 [147, 346]	1,062 [196, 2,936]	752	537	54
Smart city / traffic / OT	169	106	171 [125, 234]	2,580 [503, 5,917]	823	568	66
Call centre / help desk	74	46	690 [363, 1,160]	8,908 [3,534, 10,000]	1,685	1,391	98
All markets pooled	9,991	2,814	39 [36, 41]	149 [40, 504]	92	87	18

Table S3: Median within-year supplier HHI by product market.

Product market	Years	Count HHI	Count HHI, bias-corr.	Value HHI	Value HHI, no max
Health information systems	16	671	597	1,091	1,081
GIS / city information	12	650	369	1,503	1,206
Physical security / surveillance	4	427	108	1,098	926
Cybersecurity	7	416	156	1,666	1,263
Maintenance / support services	14	343	91	1,627	1,244
Software licences	15	325	187	1,029	868
Network / data-centre infrastructure	15	274	68	1,038	937
Computers / peripherals	15	265	44	903	762
ERP / management software	15	226	119	688	521
Custom software / web / mobile	15	176	37	743	601
All markets pooled	17	69	55	256	209

S3 Incumbency and null models

Definitions are given in the main text. Tables S4–S8 report incumbency by subgroup, each against the same permutation draws restricted to the subgroup, under N1 (market $\times$ year), N2 (adding buyer sector) and N3 (adding province). All subgroup tests under N1 have $p = 0.001$, except where noted in the main text. Table S9 reports sensitivities.

Law 7144 and procedure. Under N1, incumbency rose from 0.387 before 25 May 2018 to 0.492 afterwards (null difference 0.037). Among 21(b) contracts it rose from 0.540 to 0.810 (null difference 0.111), among open contracts from 0.385 to 0.476 (null difference 0.033). The difference-in-differences of 0.178 compares with a null of 0.078 ($p = 0.001$; $p = 0.017$ under N3). Under N3 the rise among open contracts is no longer significant ($p = 0.33$).

Buyer dependence. For the 540 buyers with at least five contracts, the share exceeding their own 95th null percentile rises with buyer size (50% for 5–9 contracts, 63% for 10–19, 80% for 20 or more), which partly reflects statistical power.

S4 Models

S4.1 Contract-level model

The primary model is reported in the main text. Robustness checks (IT-only sample, broad scope, recorded buyer labels, recorded firm names, a linear trend instead of year fixed effects, excluding 2020) give interaction odds ratios between 1.76 and 2.43. Alternative break dates give odds ratios of 2.55 (25 August 2017, KHK 694; log-likelihood -2801.1 against -2802.5 for 25 May 2018), 2.10 (placebo, May 2016, $p = 0.009$) and 1.57 (placebo, May 2020, $p = 0.08$). The interaction is 2.64 among health buyers and 0.86 ($p = 0.78$) among other buyers. A linear probability model with buyer fixed effects gives a post-2018 21(b) effect of 7.3 percentage points ($p = 0.20$), against 11.9

Table S4: Incumbency by product market.

Product market	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Health information systems	1,252	0.716	0.151 [0.136, 0.167]	0.157	0.407
Maintenance / support services	375	0.536	0.029 [0.013, 0.045]	0.139	0.122
Physical security / surveillance	182	0.418	0.030 [0.011, 0.055]	0.097	0.317
Other IT	112	0.393	0.019 [0.000, 0.045]	0.132	0.233
ERP / management software	745	0.392	0.027 [0.016, 0.039]	0.077	0.111
Network / data-centre infrastructure	328	0.384	0.023 [0.009, 0.040]	0.106	0.121
Smart city / traffic / OT	58	0.397	0.041 [0.000, 0.103]	0.104	0.314
Cybersecurity	161	0.398	0.052 [0.019, 0.087]	0.132	0.186
Computers / peripherals	281	0.327	0.012 [0.000, 0.025]	0.047	0.164
Software licences	602	0.334	0.039 [0.025, 0.055]	0.105	0.139
Education technology	80	0.350	0.068 [0.025, 0.125]	0.201	0.168
Call centre / help desk	19	0.316	0.062 [0.000, 0.158]	0.222	0.178
GIS / city information	230	0.326	0.088 [0.057, 0.122]	0.117	0.146
Custom software / web / mobile	488	0.201	0.006 [0.000, 0.014]	0.029	0.056

Table S5: Incumbency by buyer sector.

Buyer sector	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Health	1,502	0.667	0.129 [0.116, 0.142]	0.143	0.372
Other public	583	0.395	0.030 [0.017, 0.045]	0.075	0.120
Education	439	0.364	0.028 [0.014, 0.043]	0.118	0.164
Municipal/Local	1,271	0.362	0.031 [0.023, 0.041]	0.057	0.143
Provincial admin	115	0.374	0.046 [0.009, 0.087]	0.108	0.309
Social/Finance	196	0.357	0.041 [0.015, 0.066]	0.185	0.086
Agriculture/Environment	210	0.348	0.042 [0.019, 0.067]	0.110	0.100
Infrastructure/Transport	342	0.322	0.029 [0.015, 0.047]	0.097	0.116
Defence	131	0.298	0.039 [0.008, 0.076]	0.219	0.070
Security/Interior	100	0.280	0.025 [0.000, 0.060]	0.145	0.103
Justice	24	0.292	0.038 [0.000, 0.125]	0.292	0.091

($p = 0.009$) without buyer fixed effects.

S4.2 Dyad continuation

For each new buyer–firm pair in a product market, we ask whether the pair signs another contract in that market. Of 7,761 pairs, 4,334 have no later contract by the buyer in that market and cannot continue; the analysis uses the remaining 3,427, of which 34.6% continue. Covariates are measured at the first contract without look-ahead. The number of later opportunities (the buyer’s later contracts in the market) enters as a covariate; the offset restriction is rejected (coefficient 0.90 in Poisson pseudo-maximum likelihood, $p = 0.02$; 0.88 in NB2, $p = 0.006$). In NB2 the dispersion parameter is estimated ($\alpha = 0.46$, SE 0.12; likelihood-ratio statistic for $\alpha = 0$: 152.5). Firm size, firm generalism (prior markets) and prior distinct buyers are highly collinear ($r = 0.86$ and 0.99 ; VIFs above 60), so none is separately identified in the full model. Entered alone, generalism and size are both positively associated with continuation. There is no evidence that specialist firms are more likely to keep a buyer.

S5 Single bidding

Bid counts were retrieved from published result announcements for a random sample of tenders stratified by year (40 per year drawn from all retrieved tenders). Of these, 393 are in the main sample and report bid counts; none is available for 2010–2011. Rates are weighted by the inverse

Table S6: Incumbency by buyer type (keyword classes of buyer names).

Buyer type	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
MoH hospital / dental centre	617	0.831	0.177 [0.154, 0.203]	0.181	0.386
University hospital / medical faculty	110	0.627	0.099 [0.045, 0.155]	0.120	0.363
MoH provincial directorate / hospital union	695	0.563	0.098 [0.079, 0.118]	0.113	0.391
University (non-health unit)	401	0.362	0.025 [0.012, 0.040]	0.113	0.164
Municipality / municipal company	1,271	0.362	0.031 [0.023, 0.041]	0.057	0.143
Other	296	0.351	0.031 [0.014, 0.051]	0.086	0.183
Other health body (central/agency)	82	0.378	0.058 [0.012, 0.110]	0.139	0.130
Central government / agency	1,441	0.353	0.035 [0.027, 0.045]	0.124	0.107

Table S7: Incumbency by procedure.

Procedure	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Article 21(b)	618	0.723	0.143 [0.120, 0.165]	0.159	0.376
Article 21(c)	34	0.412	0.020 [0.000, 0.088]	0.069	0.255
Open	3,041	0.441	0.057 [0.051, 0.064]	0.112	0.203
Article 21(f)	1,204	0.346	0.033 [0.023, 0.043]	0.080	0.131
Article 21(a)	7	0.286	0.049 [0.000, 0.286]	0.173	0.227
Restricted	8	0.125	0.033 [0.000, 0.125]	0.059	0.035
Article 21(d)	1	0.000	0.001 [0.000, 0.000]	0.000	0.000
Article 21(e)	0	–	– [–, –]	–	–

sampling fraction of each year, with linearized 95% intervals. Incumbency status is defined as in the main text and is undefined when the tender is the buyer’s first in the market.

Adjusting for procedure and year (logit, buyer-clustered, $n = 208$ tenders with defined incumbency), the odds ratio of a single bid for incumbent winners is 4.44 (2.39–8.25). The single-bid rate has a weak upward trend (odds ratio 1.05 per year; $p = 0.04$ unweighted, $p = 0.11$ weighted). In 308 tenders with a recorded number of firms that obtained the documents, on average 4.0 firms did so and 1.9 submitted a valid bid; in 45% of these tenders two or more firms obtained the documents but only one valid bid was received.

S6 Network structure

Graph. The main-sample graph has 2,814 firms, 2,890 buyers and 7,219 weighted edges; the giant component contains 4,446 nodes (78%). Degree distributions (number of distinct partners) are heavy-tailed. Fitted by maximum likelihood with the power-law protocol of Clauset, Shalizi and Newman (2009), firms have $\hat{\alpha} = 2.21$, and a lognormal is not distinguishable from a power law ($R = -1.67$, $p = 0.09$). For buyers ($\hat{\alpha} = 2.06$) the lognormal fits better ($R = -6.19$, $p < 0.001$).

Communities. Over 50 Louvain runs with random seeds and node orders, weighted modularity of the giant component ranges from 0.760 to 0.765 (36–43 communities); Barber’s bipartite modularity of the same partitions is practically identical. In 200 bipartite degree-preserving randomizations (curveball), the unweighted modularity of the giant component is 0.694 ± 0.003 against an observed 0.731, and the weighted modularity is 0.756 ± 0.003 against an observed 0.762. Communities align with buyer sectors (NMI 0.25; permuted labels 0.03; purity 0.65 against a largest-label baseline of 0.44) and weakly with firms’ main product markets (NMI 0.13; permuted 0.04). In annual graphs, modularity exceeds its bipartite null only modestly (annual z between 0.6 and 3.8), so we do not interpret yearly modularity as evidence of persistent segmentation.

Projections and brokerage. In the bipartite configuration model (BiCM), testing all firm pairs with a false-discovery-rate correction at 0.05 validates no links, because most firms have very few buyers. Testing only co-occurring pairs yields a firm projection with 15,473 links (giant component 2,065 firms) and a buyer projection with 42,005 links. In that firm projection, betweenness correlates

Table S8: Incumbency by year.

Year	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
2026	46	0.587	0.079 [0.022, 0.152]	0.151	0.301
2025	353	0.450	0.067 [0.045, 0.093]	0.112	0.222
2024	350	0.557	0.078 [0.054, 0.106]	0.144	0.263
2023	413	0.496	0.098 [0.075, 0.121]	0.131	0.265
2022	463	0.501	0.080 [0.058, 0.102]	0.132	0.260
2021	462	0.450	0.068 [0.050, 0.089]	0.106	0.222
2020	410	0.483	0.076 [0.054, 0.100]	0.131	0.224
2019	307	0.518	0.072 [0.046, 0.101]	0.122	0.262
2018	345	0.441	0.050 [0.029, 0.072]	0.095	0.147
2017	363	0.355	0.046 [0.028, 0.066]	0.082	0.148
2016	320	0.409	0.043 [0.025, 0.066]	0.087	0.174
2015	310	0.410	0.041 [0.023, 0.065]	0.098	0.167
2014	295	0.400	0.035 [0.017, 0.054]	0.091	0.165
2013	234	0.410	0.039 [0.017, 0.064]	0.084	0.148
2012	165	0.352	0.024 [0.006, 0.048]	0.060	0.137
2011	76	0.368	0.022 [0.000, 0.053]	0.118	0.136
2010	1	0.000	0.029 [0.000, 1.000]	0.033	0.000

Table S9: Sensitivity of the incumbency result (1,000 permutations each; all $p = 0.001$).

Variant	Observed	Null mean [95% interval]	Ratio
Main (N1: market $\times$ year)	0.452	0.062 [0.056, 0.067]	7.3
N1 with 3-year windows	0.452	0.060 [0.055, 0.066]	7.5
N2: market $\times$ year $\times$ sector	0.452	0.110 [0.103, 0.116]	4.1
Buyer = recorded label	0.462	0.106 [0.100, 0.112]	4.4
Firm = recorded name	0.394	0.047 [0.042, 0.052]	8.4
Excluding call-centre contracts	0.452	0.062 [0.056, 0.067]	7.3
Including ambiguous scope	0.435	0.056 [0.051, 0.061]	7.8
Cells = buyer sector $\times$ year	0.452	0.047 [0.042, 0.052]	9.7
Incumbency at buyer level (any market)	0.389	0.053 [0.049, 0.058]	7.3
Excluding Article 21(f) contracts	0.495	0.079 [0.071, 0.087]	6.3
N3: market $\times$ year $\times$ province	0.452	0.207 [0.201, 0.213]	2.2

with the number of product markets a firm serves (Spearman 0.52) but also with its number of buyers (0.59); the partial correlation with markets given buyers and contracts is 0.12 ($p < 0.001$). Brokerage thus adds little beyond firm size. In the raw buyer projection, the most connected buyer (a provincial health directorate after splitting pooled labels) shares at least one supplier with 19% of the projection’s giant component. With recorded pooled labels the corresponding figure is 45%, which illustrates how strongly such statistics depend on buyer coding.

Table S10: Dyad continuation, logit (full specification; product-market and year fixed effects; two-way clustered 95% CIs).

	OR [95% CI]	<i>p</i>
First contract under 21(b) (vs open)	1.85 [1.37, 2.50]	0.000
First contract other negotiated (vs open)	1.26 [1.03, 1.54]	0.025
Log real value of first contract	1.02 [0.95, 1.10]	0.552
Log firm prior contracts (other buyers)	1.58 [0.97, 2.58]	0.067
Log firm prior markets (generalism)	1.20 [0.86, 1.66]	0.284
Log firm prior distinct buyers	0.69 [0.41, 1.15]	0.156
Log buyer prior contracts	0.76 [0.69, 0.85]	0.000
Log later opportunities of buyer in market	1.89 [1.68, 2.13]	0.000

Table S11: Number of later contracts, NB2 with estimated dispersion (full specification).

	IRR [95% CI]	<i>p</i>
First contract under 21(b) (vs open)	1.61 [1.39, 1.87]	0.000
First contract other negotiated (vs open)	1.19 [1.02, 1.39]	0.029
Log real value of first contract	1.01 [0.96, 1.06]	0.691
Log firm prior contracts (other buyers)	1.36 [1.07, 1.74]	0.012
Log firm prior markets (generalism)	1.20 [0.98, 1.45]	0.074
Log firm prior distinct buyers	0.76 [0.59, 0.99]	0.045
Log buyer prior contracts	0.77 [0.72, 0.82]	0.000
Log later opportunities of buyer in market	2.42 [2.23, 2.63]	0.000

Table S12: Single-bid rates (design-weighted).

Group	Tenders	Single bid, % [95% CI]	Single valid bid, %
All tenders	393	44.2 [39, 49]	55.8
Open	225	40.8 [34, 48]	55.9
Article 21(b)	50	75.0 [60, 86]	81.6
Article 21(f)	114	40.0 [31, 50]	46.9
Other negotiated	4	0.0	18.5
Winner is incumbent	95	71.2 [61, 80]	76.7
Winner is new to buyer	113	32.4 [24, 42]	48.4
Buyer's first contract in market	185	38.3 [31, 46]	50.2
Before 25 May 2018	140	38.6 [31, 47]	49.1
After 25 May 2018	253	48.9 [42, 55]	61.5